

DavarOCR: A Toolbox for OCR and Multi-Modal Document Understanding

Liang Qiao
Hikvision Research Institute
Hangzhou, China

Can Li
Hikvision Research Institute
Hangzhou, China

Baorui Zou
Hikvision Research Institute
Hangzhou, China

Yunlu Xu
Hikvision Research Institute
Hangzhou, China

Hui Jiang
Hikvision Research Institute
Hangzhou, China

Pengfei Li
Hikvision Research Institute
Hangzhou, China

Dashan Guo
Hikvision Research Institute
Hangzhou, China

Zhanzhan Cheng
Hikvision Research Institute
Hangzhou, China

Ying Chen
Hikvision Research Institute
Hangzhou, China

Zaisheng Li
Hikvision Research Institute
Hangzhou, China

Yingda Xu
Hikvision Research Institute
Hangzhou, China

Yi Niu
Hikvision Research Institute
Hangzhou, China

ABSTRACT

This paper presents DavarOCR, an open-source toolbox for OCR and document understanding tasks. DavarOCR currently implements 19 advanced algorithms, covering 9 different task forms. DavarOCR provides detailed usage instructions and the trained models for each algorithm. Compared with the previous open-source OCR toolbox, DavarOCR has relatively more complete support for the sub-tasks of the cutting-edge technology of document understanding. In order to promote the development and application of OCR technology in academia and industry, we pay more attention to the use of modules that different sub-domains of technology can share. DavarOCR is publicly released at <https://github.com/hikopenseource/Davar-Lab-OCR>.

CCS CONCEPTS

• Computing methodologies → Computer vision problems.

KEYWORDS

Open-source, OCR, Document Understanding

ACM Reference Format:

Liang Qiao, Hui Jiang, Ying Chen, Can Li, Pengfei Li, Zaisheng Li, Baorui Zou, Dashan Guo, Yingda Xu, Yunlu Xu, Zhanzhan Cheng, and Yi Niu. 2022. DavarOCR: A Toolbox for OCR and Multi-Modal Document Understanding. In *Proceedings of the 30th ACM International Conference on Multimedia (MM '22)*, October 10–14, 2022, Lisboa, Portugal. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3503161.3548547>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

MM '22, October 10–14, 2022, Lisboa, Portugal

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-9203-7/22/10...\$15.00

<https://doi.org/10.1145/3503161.3548547>

1 INTRODUCTION

OCR (Optical Character Recognition) is an essential technique in deep learning and has been applied in many fields like finance, transportation, and health care. With the continuous development of technology, people's demand for OCR ability has changed from simple text content extraction to more diversified and intelligent tasks that can directly solve the problems in actual production. Based on this, we divide the OCR tasks into two types, the *basic* OCR tasks and *document understanding* tasks.

The basic OCR tasks are the *perception* tasks that aim to obtain the content of the text in the images/videos and convert it into the electronic format, usually including text detection [1, 18, 37] and text recognition [2, 6, 25]. To be more efficient and robust, people also propose end-to-end architectures [15, 17, 22]. However, solely capturing text content sometimes cannot directly address the actual needs. In actual OCR production, intelligent systems need to perform a variety of downstream tasks based on the results of OCR. We call these tasks document understanding. These tasks can be regarded as *cognitive* tasks that require people to further understand the information (semantic information or relational information) conveyed by the text of the document. For example, Key Information Extraction (KIE) [13, 27, 35] is one of the common tasks in the financial instrument identification system, where the model needs to identify the values of the specific items like 'invoice code' and 'amount'. Other common forms of document understanding tasks include layout analysis [32, 34], Reading Order Detection (ROD) [16, 30], table recognition [14, 21] and understanding [12, 29], document Question-Answering (QA) [19], Named Entity Recognition (NER) [7, 10, 24, 31], etc.

Nowadays, many advanced open-source OCR toolboxes aim to provide more efficient and unified OCR services. Tesseract¹, chineseOCR², chineseOCR_lite³, EasyOCR⁴ are some earlier OCR

¹<https://github.com/tesseract-ocr/tesseract>

²<https://github.com/chineseocr/chineseocr>

³https://github.com/DayBreak-u/chineseocr_lite

⁴<https://github.com/JaidedAI/EasyOCR>

Table 1: Comparison of support algorithms between different open-source OCR toolboxes. All methods in Basic OCR Tasks only use visual information. In Document Understanding tasks, methods labelled with ♡ means using *visual* features, with ♣ means using *textual* features, with ♦ means using *positional* features.

Toolbox	Basic OCR Tasks				Document Understanding Tasks				
	Text Detection	Text Recognition	End-to-End Text Spotting	Video Text	KIE	NER	Layout Analysis	ROD	Table Recognition
tesseract	convention	LSTM	-	-	-	-	-	-	-
chineseocr	Yolo[23]	CRNN[25]	-	-	-	-	-	-	-
chineseocr_lite	DB[18]	CRNN[25]	-	-	-	-	-	-	-
EasyOCR	CRAFT[1]	CRNN[25]	-	-	-	-	-	-	-
PaddleOCR	DB[18], EAST[37], etc. (4)	CRNN[25], Rosetta[2], etc. (7)	PGNet[28]	-	SDMGR[27] ♡♣♦	-	Yolo[23]♡	-	RARE[26]♡
mmocr	DB[18], M-RCNN[8], etc. (6)	CRNN[25], ABINet[6], etc. (8)	-	-	SDMGR[27] ♡♣♦	BERT - Softmax[5]♣	-	-	-
davarocr	EAST[37], M-RCNN[8], TP-det[22]	CRNN[25], SPIN[33], etc. (5)	TP[22], MANGO[20], M-RCNN-e2e	YORO [4]	Chargrid[13]♡♣, TRIE[35]♡♣♦	BERT-Softmax /Span/CRF[5] ♣ BiLSTM-CRF[10]♣	VSR[34]♡♣♦	GCN-PN[16]♡♦	LGPMA[21]♡

toolboxes. They are mostly focused on providing an efficient OCR inference service and only support the tasks of text detection and recognition. PaddleOCR⁵ is a popular open-source OCR toolbox based on PaddlePaddle, which also provides relatively complete services for OCR production like labeling, training, inference and deployment. In some of its latest versions (up to v2.2), PaddleOCR starts to provide the support of some document understanding tasks, including KIE, Layout Analysis, Table Recognition and Document VQA. In each task at this stage, they provide some fairly basic and simple implementations, *e.g.*, a basic object detection solution (Yolo[23]) for layout analysis. Mmocr⁶ is a recent Pytorch-based OCR toolbox that collects many implementations of advanced OCR algorithms. It provides a rich and flexible component configuration to build an OCR model, including different backbones, necks, heads, losses, etc. Currently, mmocr has little support for document understanding tasks, *i.e.*, an KIE (SDMGR[27]) model and NER (BERT-Softmax[5]) model.

In this paper, we present DavarOCR, a new OCR toolbox that can well support the vast majority of current basic OCR and document understanding tasks. Similar to mmocr, DavarOCR is also built based on the training engine mmcv⁷ and the architecture design of mmdetection⁸. This means the whole framework can be compatible with mmocr. Therefore, we do not implement many methods for text detection and recognition algorithms but focus on expanding the tasks that MMOCR cannot support. For the basic OCR tasks, besides other text detection and recognition algorithms, we extend the framework to well support end-to-end text spotting and video text tasks. Notably, DavarOCR contains complete modules for end-to-end text spotting tasks, including both two-staged (TP[22], Mask-RCNN-based) and one-staged (MANGO[20]) text spotting architectures. For the document understanding tasks, we extend the

framework’s support for some new tasks, including Layout Analysis, ROD, and Table Recognition. Shortly, we will continue to expand the implementation of new tasks, including Document VQA and Table Understanding. At the present stage, DavarOCR contains the implementations of 19 advanced algorithms, including 10 algorithms that any other previous framework has not implemented. An algorithm-level comparison between the open-source toolboxes is shown in Table 1.

DavarOCR is released under the Apache-2.0 License. The code repository contains the one-click setup script, accompanied by the complete demo and detailed instruments to make researchers understanding easier.

2 HIGHLIGHT FEATURES OF DAVAROCR

2.1 Across Tasks Module Sharing

DavarOCR inherits the idea of modular design from mmdetection [3] and extends it to support more tasks. In mmdetection, a model can be divided into 4 four parts: BACKBONE, NECK, ROI_EXTRACTOR and HEAD. We can modify the configuration to make the model be integrated with an arbitrary combination of modules, such as different backbones. Although text detection models can mostly follow the same module divisions, other tasks will inevitably introduce new modules. In DavarOCR, some of the extended model-related modules include:

TRANSFORMATION. In text recognition, some algorithms will integrate a learnable deformation correction module to rectify the text with irregular shapes, such as Affine[11], TPS[26]. These modules can also be used in text spotting, video text tasks.

EMBEDDING. In some tasks that require textual or positional information, the model needs to utilize the features encoded by an embedding layer. For example, word/sentence embedding is used in representing the textual feature, and 2D positional embedding is used to establish spatial location connections. The related tasks include KIE, NER, Layout Analysis, ROD, etc.

⁵<https://github.com/PaddlePaddle/PaddleOCR>

⁶<https://github.com/open-mmlab/mocr>

⁷<https://github.com/open-mmlab/mmcv>

⁸<https://github.com/open-mmlab/mmdetection>

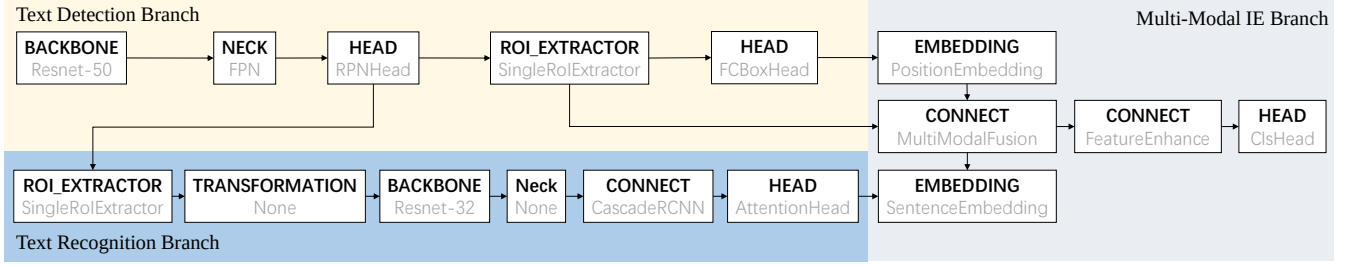


Figure 1: The modular design for the end-to-end KIE model TRIE.

CONNECT. Some intermediate connection modules are used frequently in a model like feature enhancement, feature deformation, and feature fusion. For example, Bi-LSTM[9] is a typical module used to enhance sequence features, which can be used in any text-recognition-related (text spotting, KIE, etc.) and NER tasks.

In addition to the above mentioned model-related components, many commonly used modules/operators/tools can be found in DavarOCR. All of the above modular design in DavarOCR allows for maximum component sharing between different tasks, enabling researchers to build a cross-task model quickly. Figure 1 demonstrates an example architecture of the modal TRIE[35], which is an end-to-end model contains three sub-tasks: text detection, recognition and information extraction.

2.2 Uniform Data Label Format

Because of the different types of tasks, we often need to use different supervision to train models, which leads to varying forms of task-dependent data annotation. In fact, there are many similarities between basic OCR and document understanding tasks. To further unify the data processing logic in DavarOCR, We try to integrate all possible tasks and propose some unified basic formats for the data label.

```
{
  "Images/train/img1.jpg": {
    "height": int,      # Image height
    "width": int,       # Image width
    "content_ann": {    # First-level annotations
      "bboxes": List[List[int|float]], # Bounding box for each obj
      "cbboxes": List[List[List[int|float]]], # Character-wised bounding boxes
      "cares": List[int], # Whether the bboxes is ignored
      "labels": List[List[int]], # Category for each box
      "texts": List[str] # Transcriptions of text
    },
    "content_ann2": {   # Second-level annotations
      ...               # Same format with 1st-level
    }
  },
  ...
}
```

Figure 2: Illustration of the basic data label

Figure 2 shows an illustration of the basic image-based data label format. In the basic format, annotation information for all images is arranged in a JSON file, and the keys are the images' paths. The detailed annotation is stored in the item of "content_ann". All the items in "content_ann" are array types and have the same length on the outermost level. This indicates that all the labeling

information corresponds to each bounding box. If someone needs to store full-image-level annotations, all the lists are of length 1, and "bboxes" default to contain a single box to represent the entire image range. Item of "labels" saves the category information for an object or the whole image, which is used in the tasks related to object classification like KIE and Layout Analysis. It is stored in a two-dimensional list to support the classification of multitasking. Notably, we add keywords ("content_ann2") to save multilevel labels for supporting the multilevel task. For example, in the multi-modal layout analysis task, the model needs to simultaneously learn the page-level (block-level) and text-line-level tasks.

The basic data label format supports most image-based tasks, including text detection, text recognition, end-to-end text spotting, KIE, and Layout Analysis. When we use the data to train a model, we can freely select any combination of the keywords to form the required information for the task. We can define the task-specific data form for new tasks that inherit the basic format to supplement the required items. For example, in table recognition tasks, the model relies on the item to represent cell indexes. DavarOCR also provides the data format for the video-based (Video Text) of plain-text-based (NER) tasks.

Table 2: Multi-modal ablation results of the Key Information Extraction task on WildReceipt[27].

Algorithm	Modalities	Backbone	Testing Scale	F1-Score
Chargrid[13]	Visual	ChargridNet	(512,512)	67.10
	Visual+Textual	ChargridNet	(512,512)	81.02
TRIE[35]	Visual	ResNet-50	(512,512)	78.25
	Visual+Textual	ResNet-50	(512,512)	84.67
	Visual+Positional	ResNet-50	(512,512)	86.57
	Visual+Textual+Positional	ResNet-50	(512,512)	87.08

Through the uniform data format, most of the tasks from different domains can share the same data processing functions like data loading, data augmentation and data formatting.

Table 3: Multi-modal ablation results of the Layout Analysis task on PubLayNet[36]. (C) means Character-level textual feature, and (S) means Sentence-level textual feature.

Algorithm	Modalities	Backbone	Testing Scale	mAP
VSR[34]	Visual	ResNeXt-101	(1300,800)	92.6
	Visual+Textual (C)	ResNeXt-101	(1300,800)	94.5
	Visual+Textual (S)	ResNeXt-101	(1300,800)	94.7
	Visual+Textual (C+S)+Positional	ResNeXt-101	(1300,800)	95.8

3 CASE STUDY: MULTI-MODAL DOCUMENT UNDERSTANDING

Different modal information may play different roles in document understanding tasks. This section demonstrates some of the modality ablation results in DavarOCR.

Table 2 and Table 3 separately show some models' ablation results on the KIE and Layout Analysis tasks. In the KIE experiments, the task can be simply implemented by a single visual-based model, which equals an object detection task. The Chargrid model uses a char-level segmentation map to introduce textual features, and TRIE will extract the positional and textual features to compromise the box-level multi-modal features. From the results, we can see that involving additional modalities can greatly improve the model's performance.

4 CONCLUSION

This paper presents an open-source toolbox, DavarOCR, for comprehensive OCR and document understanding tasks. DavarOCR currently implements 19 state-of-the-art algorithms, covering 9 different task areas. In DavarOCR, researchers can freely combine the reusable modules and adopt multi-modal information to implement more complex algorithms involving perception and cognition. We hope that this framework can further promote the implementation of new technologies in actual production.

REFERENCES

- [1] Youngmin Baek, Bado Lee, Dongyoon Han, Sangdoo Yun, and Hwalsuk Lee. 2019. Character region awareness for text detection. In *CVPR*. 9365–9374.
- [2] Fedor Borisov, Albert Gordo, and Viswanath Sivakumar. 2018. Rosetta: Large Scale System for Text Detection and Recognition in Images. In *KDD*. 71–79.
- [3] Kai Chen, Jiaqi Wang, Jiangmiao Pang, Yuhang Cao, Yu Xiong, Xiaoxiao Li, Shuyang Sun, Wansen Feng, Ziwei Liu, Jiarui Xu, Zheng Zhang, Dazhi Cheng, Chenchen Zhu, Tianheng Cheng, Qijie Zhao, Buyu Li, Xin Lu, Rui Zhu, Yue Wu, Jifeng Dai, Jingdong Wang, Jianping Shi, Wanli Ouyang, Chen Change Loy, and Dahua Lin. 2019. MMDetection: Open MMLab Detection Toolbox and Benchmark. *CoRR* abs/1906.07155 (2019).
- [4] Zhanzhan Cheng, Jing Lu, Yi Niu, Shiliang Pu, Fei Wu, and Shuigeng Zhou. 2019. You Only Recognize Once: Towards Fast Video Text Spotting. In *ACM MM*. ACM, 855–863.
- [5] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *NAACL-HLT*. 4171–4186.
- [6] Shancheng Fang, Hongtao Xie, Yuxin Wang, Zhendong Mao, and Yongdong Zhang. 2021. Read Like Humans: Autonomous, Bidirectional and Iterative Language Modeling for Scene Text Recognition. In *CVPR*. 7098–7107.
- [7] Jiafeng Guo, Gu Xu, Xueqi Cheng, and Hang Li. 2009. Named entity recognition in query. In *SIGIR*. 267–274.
- [8] Kaiming He, Georgia Gkioxari, Piotr Dollar, and Ross Girshick. 2017. Mask R-CNN. In *ICCV*. 2980–2988.
- [9] Sepp Hochreiter and Jürgen Schmidhuber. 1997. Long Short-Term Memory. *Neural Computation* 9, 8 (1997), 1735–1780.
- [10] Zhiheng Huang, Wei Xu, and Kai Yu. 2015. Bidirectional LSTM-CRF Models for Sequence Tagging. *CoRR* (2015).
- [11] Max Jaderberg, Karen Simonyan, Andrew Zisserman, and Koray Kavukcuoglu. 2015. Spatial Transformer Networks. In *NeurIPS*. 2017–2025.
- [12] Sujay Kumar Jauhar, Peter D. Turney, and Eduard H. Hovy. 2016. Tables as Semi-structured Knowledge for Question Answering. In *ACL*.
- [13] Anoop R. Katti, Christian Reisswig, Cordula Guder, Sebastian Brarda, Steffen Bickel, Johannes Höhne, and Jean Baptiste Faddoul. 2018. Chargrid: Towards Understanding 2D Documents. In *EMNLP*. 4459–4469.
- [14] Saqib Ali Khan, Syed Muhammad Daniyal Khalid, Muhammad Ali Shahzad, and Faisal Shafait. 2019. Table Structure Extraction with Bi-Directional Gated Recurrent Unit Networks. In *ICDAR*. 1366–1371.
- [15] Hui Li, Peng Wang, and Chunhua Shen. 2017. Towards End-to-end Text Spotting with Convolutional Recurrent Neural Networks. In *ICCV*. 5248–5256.
- [16] Liangcheng Li, Feiyu Gao, Jiajun Bu, Yongpan Wang, Zhi Yu, and Qi Zheng. 2020. An End-to-End OCR Text Re-organization Sequence Learning for Rich-Text Detail Image Comprehension. In *ECCV*, Vol. 12370. 85–100.
- [17] Minghui Liao, Pengyuan Lyu, Minghang He, Cong Yao, Wenhao Wu, and Xiang Bai. 2019. Mask textspotter: An end-to-end trainable neural network for spotting text with arbitrary shapes. *IEEE TPAMI* 1, 1 (2019).
- [18] Minghui Liao, Zhaoyi Wan, Cong Yao, Kai Chen, and Xiang Bai. 2020. Real-Time Scene Text Detection with Differentiable Binarization. In *AAAI*. 11474–11481.
- [19] Minesh Mathew, Dimosthenis Karatzas, and C. V. Jawahar. 2021. DocVQA: A Dataset for VQA on Document Images. In *WACV*. 2199–2208.
- [20] Liang Qiao, Ying Chen, Zhanzhan Cheng, Yunlu Xu, Yi Niu, Shiliang Pu, and Fei Wu. 2021. MANGO: A Mask Attention Guided One-Stage Scene Text Spotter. In *AAAI*. 2467–2476.
- [21] Liang Qiao, Zaisheng Li, Zhanzhan Cheng, Peng Zhang, Shiliang Pu, Yi Niu, Wenqi Ren, Wenming Tan, and Fei Wu. 2021. LGPMA: Complicated Table Structure Recognition with Local and Global Pyramid Mask Alignment. In *ICDAR*, Vol. 12821. 99–114.
- [22] Liang Qiao, Sanli Tang, Zhanzhan Cheng, Yunlu Xu, Yi Niu, Shiliang Pu, and Fei Wu. 2020. Text Perceptron: Towards End-to-End Arbitrary-Shaped Text Spotting. In *AAAI*. 11899–11907.
- [23] Joseph Redmon and Ali Farhadi. 2017. YOLO9000: Better, Faster, Stronger. In *CVPR*. IEEE Computer Society, 6517–6525.
- [24] Erik F. Tjong Kim Sang and Fien De Meulder. 2003. Introduction to the CoNLL-2003 Shared Task: Language-Independent Named Entity Recognition. In *CoNLL/HLT-NAACL*. 142–147.
- [25] Baoguang Shi, Xiang Bai, and Cong Yao. 2017. An End-to-End Trainable Neural Network for Image-Based Sequence Recognition and Its Application to Scene Text Recognition. *IEEE TPAMI* 39, 11 (2017), 2298–2304.
- [26] Baoguang Shi, Xinggang Wang, Pengyuan Lyu, Cong Yao, and Xiang Bai. 2016. Robust Scene Text Recognition with Automatic Rectification. In *CVPR*. 4168–4176.
- [27] Hongbin Sun, Zhanghui Kuang, Xiaoyu Yue, Chenhao Lin, and Wayne Zhang. 2021. Spatial Dual-Modality Graph Reasoning for Key Information Extraction. *CoRR* abs/2103.14470 (2021).
- [28] Pengfei Wang, Chengquan Zhang, Fei Qi, Shanshan Liu, Xiaoqiang Zhang, Pengyuan Lyu, Junyu Han, Jingtuo Liu, Errui Ding, and Guangming Shi. 2021. PGNet: Real-time Arbitrarily-Shaped Text Spotting with Point Gathering Network. In *AAAI*. 2782–2790.
- [29] Zhiruo Wang, Haoyu Dong, Ran Jia, Jia Li, Zhiyi Fu, Shi Han, and Dongmei Zhang. 2020. Structure-aware Pre-training for Table Understanding with Tree-based Transformers. *CoRR* abs/2010.12537 (2020).
- [30] Zilong Wang, Yiheng Xu, Lei Cui, Jingbo Shang, and Furu Wei. 2021. LayoutReader: Pre-training of Text and Layout for Reading Order Detection. In *EMNLP*. 4735–4744.
- [31] Hang Yan, Tao Gui, Junqi Dai, Qipeng Guo, Zheng Zhang, and Xipeng Qiu. 2021. A Unified Generative Framework for Various NER Subtasks. In *ACL*. 5808–5822.
- [32] Xiao Yang, Ersin Yumer, Paul Asente, Mike Kraley, Daniel Kifer, and C. Lee Giles. 2017. Learning to Extract Semantic Structure from Documents Using Multimodal Fully Convolutional Neural Networks. In *CVPR*. 4342–4351.
- [33] Chengwei Zhang, Yunlu Xu, Zhanzhan Cheng, Shiliang Pu, Yi Niu, Fei Wu, and Futa Zou. 2021. SPIN: Structure-Preserving Inner Offset Network for Scene Text Recognition. In *AAAI*. AAAI Press, 3305–3314.
- [34] Peng Zhang, Can Li, Liang Qiao, Zhanzhan Cheng, Shiliang Pu, Yi Niu, and Fei Wu. 2021. VSR: A Unified Framework for Document Layout Analysis Combining Vision, Semantics and Relations. In *ICDAR*, Vol. 12821. 115–130.
- [35] Peng Zhang, Yunlu Xu, Zhanzhan Cheng, Shiliang Pu, Jing Lu, Liang Qiao, Yi Niu, and Fei Wu. 2020. TRIE: End-to-End Text Reading and Information Extraction for Document Understanding. In *ACM MM*. ACM, 1413–1422.
- [36] Xu Zhong, Jianbin Tang, and Antonio Jimeno-Yepes. 2019. PubLayNet: Largest Dataset Ever for Document Layout Analysis. In *ICDAR*. IEEE, 1015–1022.
- [37] Xinyu Zhou, Cong Yao, He Wen, Yuzhi Wang, Shuchang Zhou, Weiran He, and Jiajun Liang. 2017. EAST: An Efficient and Accurate Scene Text Detector. In *CVPR*. 2642–2651.